



examGenome CheatSheet on Algebra

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1	What is the main formula used in 'Rate of Work' problems? Answer: $\text{Work} = \text{Rate} \times \text{Time}$, but usually we use: $\text{Rate} = 1 \div \text{Time}$ (for one whole job). Explanation: Since the whole job is treated as '1', if someone takes 5 hours to finish, their rate is $\frac{1}{5}$ of the job per hour.
2	What are the units used for 'Rate of Work'? Answer: Jobs per hour, tasks per minute, or pools per day—always 'amount of work per unit of time'. Explanation: For example: '$\frac{1}{3}$ of a room per hour' or '2 cakes per hour'. The unit is always work divided by time.
3	What does '1 job' mean in rate of work? Answer: It means one complete task—like cleaning one room, painting one fence, or filling one tank. Explanation: We pretend the whole task is equal to the number 1 so we can use fractions and simple math.
4	What is the rule for combining two people's work rates? Answer: Add their individual rates: $\text{Combined Rate} = \text{Rate}_A + \text{Rate}_B$. Explanation: If you do $\frac{1}{4}$ of the job per hour and your friend does $\frac{1}{4}$ per hour, together you do $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ per hour.
5	Is time directly or inversely proportional to the number of workers? Answer: Inversely proportional—more workers mean less time (if they work at the same speed). Explanation: Doubling the number of workers (with same skill) usually halves the time needed.
6	Can a work rate be greater than 1? Answer: Yes, if someone can do more than one whole job in one hour (e.g., baking 3 cakes in 1 hour $\rightarrow$ rate = 3 cakes/hour). Explanation: Rates can be fractions (less than 1) or whole numbers (1 or more), depending on how fast the work is done.
7	What is the key property of work rates when people work together? Answer: Their rates add up, but only if they work independently and at the same time. Explanation: This only works if both are doing the same kind of task and don't slow each other down.
8	Why do we use fractions for work rates? Answer: Because most people take more than 1 hour to finish a job, so they do only a part (fraction) of it each hour. Explanation: Fractions help us show 'how much' of the job is done in one unit of time.
9	What assumption do we always make in simple rate of work problems? Answer: Everyone works at a steady, constant speed, and the job can be split into equal parts. Explanation: We assume no breaks, no tiredness, and that the work can be shared evenly (like painting a wall).

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10	How is 'Rate of Work' different from 'Speed'? Answer: Speed is about distance over time (km/h), while Rate of Work is about jobs over time (jobs/hour). Explanation: Both use the idea of 'how much per hour', but one is for moving, the other is for completing tasks.

